AZHAR AHMED

Data Scientist | AI/ML Developer | IT Undergraduate

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PROFESSIONAL SUMMARY`

Certified Data Scientist and Artificial Intelligence enthusiast currently pursuing a BS in Information Technology at the University of Sindh. Skilled in Machine Learning, Deep Learning, Data Analysis, and Python development, with hands-on experience building predictive models, neural networks, and interactive data applications. Passionate about leveraging data-driven technologies to solve real-world problems and continuously expanding expertise through practical projects and professional certifications.

EDUCATION

Bachelor of Science in Information Technology 2025 – 2029

University of Sindh, Jamshoro

FSc. Pre-Engineering 2022 – 2024

Federal Govt. Degree College, Cantt

TECHNICAL SKILLS

Programming Languages	Python • Java • C++ • SQL • HTML • CSS • JavaScript
Data Analysis	NumPy • Pandas
Machine Learning	Scikit-learn • XGBoost • LightGBM • Regression, Classification & Clustering • Feature Engineering • Hyperparameter Tuning (GridSearchCV)
Deep Learning	TensorFlow • Keras • PyTorch • ANN • CNN • RNN / LSTM • Transfer Learning
Natural Language Processing (NLP)	Hugging Face Transformers • BERT • Word Embeddings
Generative AI & LLMs	Prompt Engineering • LangChain • Retrieval-Augmented Generation (RAG) • APIs • Fine-Tuning
Computer Vision	OpenCV • Image Classification & Object Detection • CNN Architectures (ResNet, VGG, YOLO)
Data Visualization	Matplotlib • Seaborn • Plotly
Tools & Platforms	Jupyter Notebook • Google Colab • Anaconda • Git • GitHub • VS Code
Core Competencies	Predictive Modeling • Data Wrangling & Cleaning • Exploratory Data Analysis (EDA) • Model Evaluation & Validation • Problem Solving

PROJECTS

FasalGuard — Crop Disease Detection App | *Python |FastApi | YOLOv8 | ViT | React Native | Hugging Face | Groq*

- Designed and deployed a two-stage computer vision system that detects and classifies crop diseases from leaf photos, achieving 93% accuracy across 38 disease types and 13 crop species.
- Shipped as a full-stack mobile product (React Native + FastAPI) with an AI chatbot for treatment advice, giving farmers an end-to-end diagnosis-to-action tool.

- Fine-tuned a Vision Transformer (ViT) on 87,000+ images from PlantVillage to classify 38 disease classes across 13 crop species, achieving 93% classification accuracy.

PneumoDetect — Pneumonia Detection from Chest X-Rays | *Python | Flask | PyTorch | Vision Transformer (ViT)*
- Built a web application that uses a fine-tuned Vision Transformer to detect pneumonia from chest X-rays in real time, with confidence scores to support clinical decision-making.
- Optimized the model architecture for speed and low memory use, then containerized the app with Docker for reliable deployment.

SpamShield — Email Spam Detection Using NLP | *Python | Flask | DistilBERT | Hugging Face Transformers | PyTorch*
- Trained a transformer-based NLP model on 190,000+ emails to accurately flag spam, capturing subtle language patterns that simpler filters miss.
- Deployed as a web app with automatic GPU acceleration, enabling fast, real-time predictions.

Heart Disease Prediction Using Machine Learning | *Python | Pandas | Scikit-learn | Matplotlib | Seaborn*
- Built and compared multiple machine learning models to predict cardiovascular disease risk from patient health data, tuning for recall to minimize missed diagnoses. A key priority in healthcare applications.

California Housing Price Prediction Using Artificial Neural Networks | *Python | TensorFlow | Keras | NumPy | Scikit-learn*
- Developed a neural network model to predict housing prices, using regularization techniques to keep predictions reliable on new, unseen data.

Fashion MNIST Image Classification Using Convolutional Neural Networks | *Python | TensorFlow | Keras*
- Built a convolutional neural network that classifies clothing images with 90%+ accuracy across 10 categories, demonstrating core computer vision fundamentals.

Interactive Data Analysis & Visualization Dashboard | *Python | Streamlit | Pandas | NumPy | Matplotlib | Seaborn*
- Built a Streamlit-based EDA web app that lets users upload datasets and instantly generate descriptive statistics, interactive visualizations, missing-value reports, and downloadable summaries, turning manual data exploration into a self-service tool for non-technical users.

CERTIFICATIONS

- Artificial Intelligence & Data Science Training Program (14 months) — Saylani Mass IT Training
- Introduction to Modern AI — Cisco Networking Academy (NetAcad)
- PITP Certified Java Developer — People's IT Training Programme (PITP)
- PITP Certified Web Developer — People's IT Training Programme (PITP)

AREAS OF INTEREST

‣ Machine Learning & Deep Learning	‣ Data Science & Predictive Analytics
‣ Artificial Intelligence Applications	‣ Data Visualization & Business Intelligence
‣ Computer Vision	‣ AI-Powered Software Development

LANGUAGES

English — Professional Working Proficiency
Urdu — Native